

StockVision – Problem Statement

Problem Statement | Scilab GUIVerse Hackathon

1. The Problem

I saw a gap between introductory Scilab examples and an interactive application that lets a learner explore a real predictive workflow. Stock-price prediction is often presented as a black box: data goes in and a number comes out, while the user has little opportunity to see how different modelling approaches, training history, validation choices, and practical assumptions affect the result.

I chose stock-market modelling because it naturally combines data visualization, forecasting, comparison, validation, and decision-oriented analysis in one GUI workflow. It also fits the hackathon's focus on original GUI applications, interactive problem solving, educational usefulness, and applications that go beyond basic calculators and plotting examples.

2. The Gap I Wanted to Fill

I wanted a lightweight Scilab application where a user could answer questions such as:

- How does a feature-based Linear Regression prediction compare with an autoregressive time-series prediction on the same data?
- How does a smoothing-based trend model such as Holt's linear trend method compare with those two approaches?
- What changes when the train/test split is moved to a different portion of the available history?
- Does a 20-day moving average support or contradict the trend identified by the models?
- Do the three models remain useful under forward-only walk-forward validation rather than a single fixed split?
- What happens when a model-guided signal is evaluated with a one-day execution lag, transaction costs, and slippage?

Normally, answering these questions requires writing or modifying code, running separate experiments, and comparing outputs manually. I wanted to turn them into a single interactive GUI workflow.

3. What I Built

I built StockVision as an interactive Scilab GUI that fits three different statistical approaches to OHLCV data: feature-based Linear Regression, a configurable autoregressive time-series model, and Exponential Smoothing using Holt's linear trend method. The user selects a dataset and model, changes meaningful parameters, runs the analysis, compares all three approaches, validates them forward in time, backtests the resulting signal, and exports the results.

- Compare three forecasting approaches on the same dataset and train/test split.
- Change the train/test split interactively with a slider.
- Use Model Info to understand the currently selected model in plain language.
- Inspect RMSE, MAE, MAPE, R-squared, and Prediction Accuracy.
- View a next-day estimate, rough uncertainty band, and colour-coded BUY/SELL/HOLD signal.
- Run 3-fold walk-forward validation across all three models.
- View a dedicated comparison chart of single-split and walk-forward RMSE.
- Backtest model-guided signals against Buy & Hold with a one-day execution lag.
- Include transaction cost and slippage assumptions and report risk metrics.

4. Intended Users and Use Cases

I designed StockVision primarily for students and learners studying regression, time-series modelling, smoothing methods, data visualization, and basic financial analysis. A user can compare different modelling philosophies, observe how parameter choices affect results, understand evaluation metrics, and see how a simple historical strategy behaves under explicit assumptions.

5. Why I Chose This Approach

I deliberately chose depth of interaction over breadth of features. Each major GUI element changes a real modelling, validation, signal, or backtesting input. The three-model design lets the user compare three different ways of representing the past: engineered features, a fixed autoregressive lookback, and a running level-and-trend state.

The application uses a substantial range of Scilab GUI concepts rather than only the minimum requirement, including menus, popup menus, sliders, checkboxes, edit controls, push buttons, frames, results controls, message boxes, dedicated chart windows, and file dialogs.

6. Scope and Limitations

The bundled datasets are synthetic demonstration data, not live or downloaded historical prices. The application has no live market-data connection. It is an educational and analytical tool, not investment advice or a guarantee of market prediction or profitability.

The backtest uses a simplified execution model with a one-day signal-to-trade lag and configurable transaction cost/slippage percentages. It does not model bid/ask spread, partial fills, intraday movement, or market impact, and the strategy is long/cash only. Exponential Smoothing uses fixed smoothing parameters rather than fitting them from the data.

7. Expected Outcome

My objective is not to claim that one model can reliably predict financial markets. My objective is to make the modelling process understandable and testable: users should be able to compare three modelling philosophies, question the effect of parameter choices, validate performance forward in time, and see how a simple strategy behaves under explicit assumptions.

In one sentence: I built StockVision to turn stock forecasting from a single black-box prediction into an interactive Scilab workflow for model comparison, validation, visualization, and practical evaluation.